LESSON PLAN

EHM (28)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to common measuring equipment - Vernier Caliper, Screw Gauge, Feeler Gauge
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to common measuring equipment - Vernier Caliper, Screw Gauge, Feeler Gauge
5. Specific Objectives	28.1.2.3 Trainee will be able to explain correct usages common measuring equipment.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of zero error correction and reading techniques on Vernier Caliper, Screw Gauge, and Feeler Gauge; Safety: Handle jaws gently to avoid damage; Instructor supervision: Monitor trainee handling of precision instruments.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the standard unit of length used in aviation engineering?
(b) Why is precision critical in aircraft maintenance?
(c) Can you name three types of measuring tools used for linear dimensions?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Vernier Caliper construction and parts • Reading main scale and vernier scale • Correct usage of jaws and depth probe • Zero error identification and correction • Applications in aircraft component measurement	L & Ds	PPT, IB	15 min
(b)	• Screw Gauge (Micrometer) mechanism • Thimble and sleeve scale reading • Zero error calculation for micrometers • Proper handling to prevent thread damage • Use in measuring small diameters and thickness	L & Ds	PPT, PDF	20 min
(c)	• Feeler Gauge blade identification • Selection of correct gap thickness • Insertion technique for clearance checks • Common applications in valve and gap measurement • Care and storage of feeler gauge sets	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a Vernier Caliper and a Screw Gauge in terms of measurement range.
(b) How do you correct a positive zero error on a Screw Gauge?
(c) When would you choose a Feeler Gauge over a Vernier Caliper for a maintenance task?

12. Home Assignment and Follow Up	Prepare a brief note describing the correct procedure for checking zero error on a Vernier Caliper before taking any measurement, referencing the IB guidelines.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Types of Wire locking, Practical on wire locking procedure	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min